



Knowledge Runner

Transforming active recall through generative AI and
engaging gameplay

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Problem: The Active Recall Deficit

Ineffective Study Habits

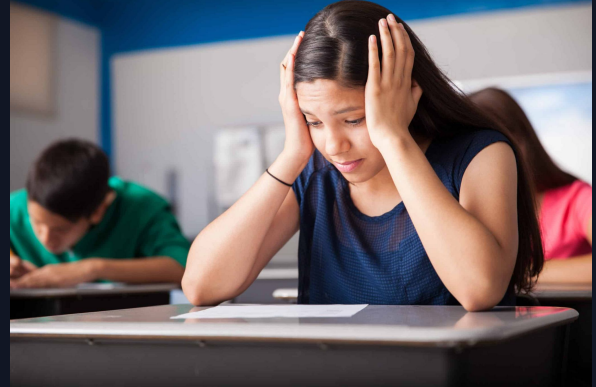
- **84%** of college students rely on re-reading and highlighting.
- Re-reading builds **familiarity, not retrieval strength.**

The Testing Effect Deficit

- Active retrieval improves long-term retention by **400%+** compared to repeated studying

Digital Attention Fragmentation

- Average screen focus before switching: only **47 seconds.**



The Solution

- **Infinite Curriculum:** Type any topic (AI Ethics, French Vocab, Organic Chemistry)

KNOWLEDGE RUNNER

QUIZ TOPIC

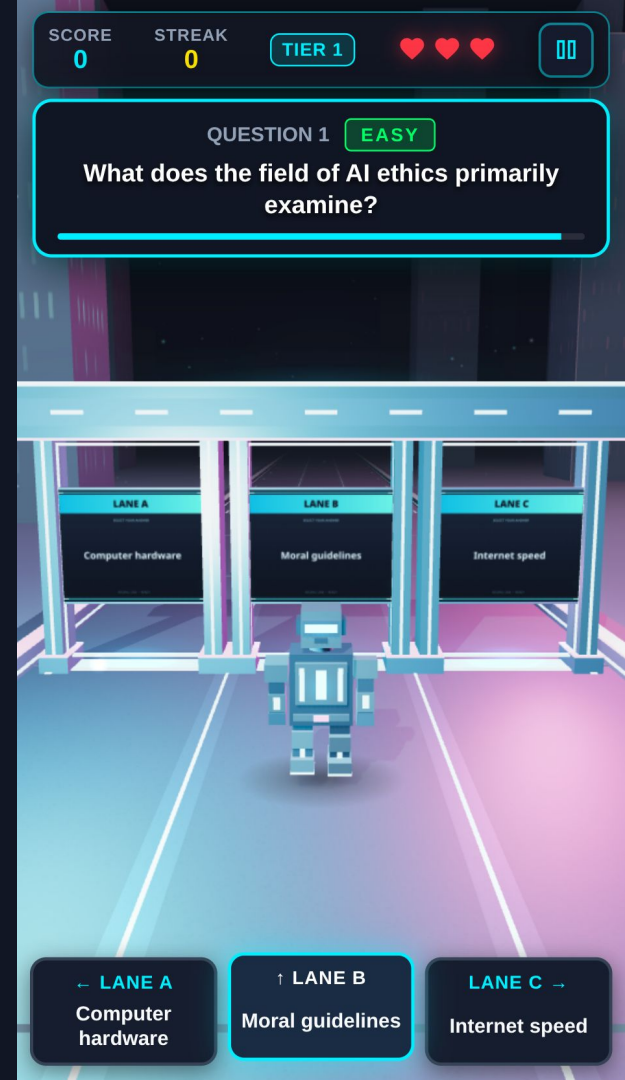
AI Ethics

START

Controls: Left/Right Arrow Keys, A / D, or tap
bottom lane cards.

The Solution

- **Fun Gameplay:** Steer left/center/right to run through the correct answer gate
- **Risk / Reward:** Streak multiplier and +1 life recovery on streaks



The Solution

- **Revision Loop:** Full game-over review screen showing chosen vs. correct answers with explanations

RUN COMPLETE

Here is your full revision summary:

FINAL SCORE QUESTIONS BEST STREAK

600 4 / 7 4

5. What does 'value alignment' mean in AI development?

✗ You chose: Reducing data |
Correct: Matching human goals

Value alignment is the effort to ensure AI systems act in ways consistent with human values and intended outcomes.

6. Who ultimately bears responsibility for harmful AI outcomes?

✗ You chose: Random users |
Correct: Human developers

RUN AGAIN

NEW TOPIC

SDG Goal 4: Quality education

- **Target 4.4:** Relevant cognitive skills for academic and vocational success.
- **Target 4.7:** Accessible, self-directed educational technology that removes the barrier of expensive private tutoring.



4 QUALITY
EDUCATION



Target User Group & Scope

- **Primary Users:** High school / college students (15–24) and self-learners revising technical topics
- **Context of Use:** 3–5 minute high-intensity review sessions on mobile or laptop
- **User Needs:** Instant startup, zero manual deck writing
- **Excluded Users:**
 - Under-10s (game requires rapid 1–3 word reading at high speed)
 - Long-form essay writers (multiple-choice format is unfit)

Problem–Solution Fit

Why this beats traditional alternatives:

- **Vs. Flashcards**

Game-flow urgency → less boredom & passive skimming.

- **Vs. Kahoot**

Unlimited content → never runs out of questions.

- **Vs. LLM Chatbots**

Instant retrieval under time pressure → faster than text-based chats.

AI Tool Integration: Built with Bolt.new



- **Architecture Generation:** Leveraged Bolt.new to write modular JS, 3D graphics, and audio
- **Resilient API Handling:** Prompted Bolt to generate strict JSON parsing with auto-retry logic for malformed LLM responses
- **Live Sandbox:** Developed and debugged the full engine in Bolt's sandbox
- **Why Bolt.new Was Essential:** Reduced days of complex WebGL, state-machine, and prompt pipeline writing

Ethical Reasoning & Risk Mitigation

- **Risk: Hallucinated Answers**

- Impact: Reinforcing false facts during fast gameplay
- Fix: Strict prompt rules, low LLM temperature (0.7), and post-run explanations; roadmap includes RAG grounding

- **Risk: Cognitive Overload**

- Impact: Visual distraction hurting reading comprehension
- Fix: Strict 1–3 word option limits + high-contrast bottom HUD cards for easy reading

- **Risk: Cultural / Model Bias**

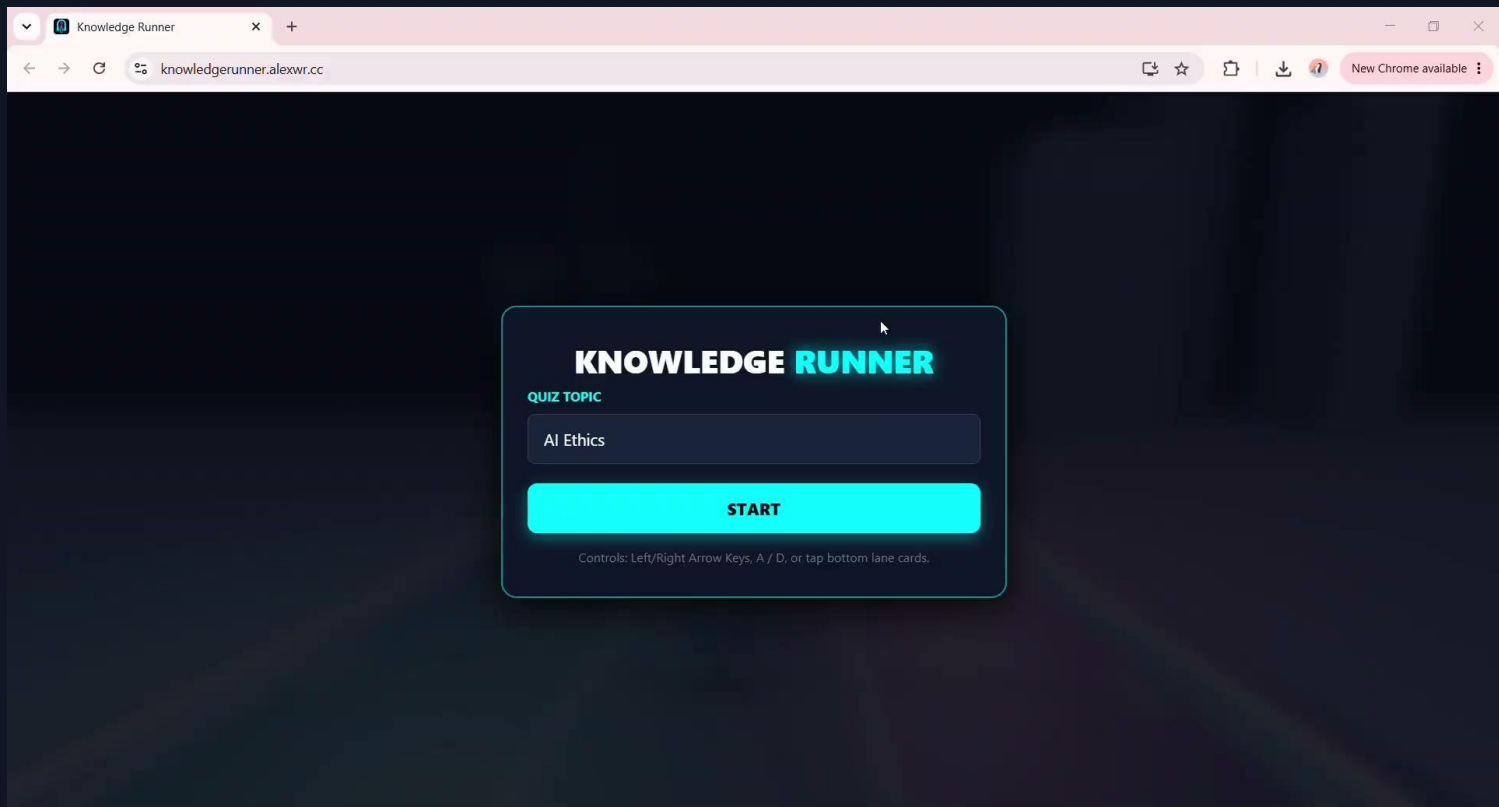
- Impact: Subjective trivia framed as absolute fact
- Fix: Detailed post-run explanation logs rather than unexplained "wrong" markers

What does it not do?

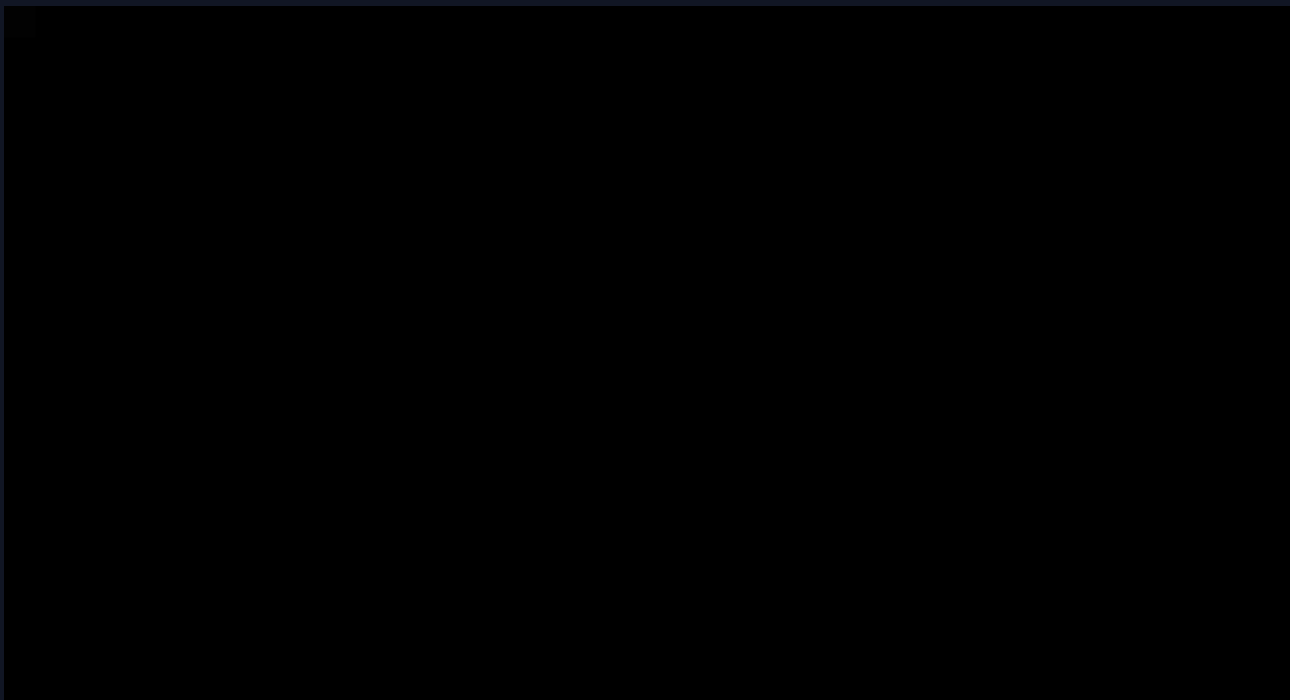
- **No accounts:** No logins, profiles, or tracking.
- **No progress saving:** Scores and history reset after refresh.
- **No classroom tools:** No teacher dashboards, LMS integration, or grade exports.



Demo



Demo



Try it out!



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